Study 3: Identity fusion, social identity, and expected moral credit

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Configuration

```
data_path <- "/Users/yaldasmac/Desktop/Metaperception/Study\ 3/study3_data.xlsx"

lib_code  <- 1   # Group == 1: liberals
con_code  <- 2   # Group == 2: conservatives
cond_out  <- 1   # Condition == 1: outgroup perception (benchmark)
cond_meta <- 2   # Condition == 2: metaperception

binding_items <- c("loyalty_1", "loyalty_2", "authority_1", "authority_2",
                   "purity_1", "purity_2", "proportionality_1", "proportionality_2")
indiv_items   <- c("care_1", "care_2", "equality_1", "equality_2")

identity_items <- paste0("identity_", 1:6)
fusion_col     <- "fusion"
ideology_col   <- "ideology"
```

```
library(readxl)
library(broom)
library(car)
library(effectsize)
library(knitr)
```

```
fmt_p <- function(p) ifelse(p < .001, "< .001", sub("^0", "", sprintf("%.3f", p)))

# Coefficient table with CI, standardized beta, and Type III partial eta2
coef_table <- function(model, label) {
  tt  <- tidy(model, conf.int = TRUE)
  tt  <- tt[tt$term != "(Intercept)", ]
  mf  <- model.frame(model)
```

```
  sdy <- sd(mf[[1]])
  df  <- df.residual(model)
  data.frame(
    model   = label,
    term    = tt$term,
    b       = tt$estimate,
    SE      = tt$std.error,
    CI_low  = tt$conf.low,
    CI_high = tt$conf.high,
    beta    = tt$estimate * sapply(tt$term, function(v) sd(mf[[v]])) / sdy,
    t       = tt$statistic,
    df      = df,
    p       = fmt_p(tt$p.value),
    eta2_p  = tt$statistic^2 / (tt$statistic^2 + df),   # Type III partial eta2
    row.names = NULL
  )
}

fit_summary <- function(model, label) {
  s <- summary(model)
  f <- s$fstatistic
  data.frame(model = label, n = nobs(model),
             F = f[["value"]], df1 = f[["numdf"]], df2 = f[["dendf"]],
             p = fmt_p(pf(f[["value"]], f[["numdf"]], f[["dendf"]], lower.tail = FALSE)),
             R2 = s$r.squared, adj_R2 = s$adj.r.squared)
}
```

1. Data and composites

```
data <- read_excel(data_path)

needed <- c(binding_items, indiv_items, identity_items, fusion_col,
            ideology_col, "Group", "Condition")
miss <- setdiff(needed, names(data))
if (length(miss)) stop("Missing columns: ", paste(miss, collapse = ", "))

data$binding         <- rowMeans(data[, binding_items], na.rm = TRUE)
data$individualizing <- rowMeans(data[, indiv_items],   na.rm = TRUE)
data$identity        <- rowMeans(data[, identity_items], na.rm = TRUE)
data$fusion          <- as.numeric(data[[fusion_col]])
data$ideology        <- as.numeric(data[[ideology_col]])

lib_meta <- subset(data, Group == lib_code & Condition == cond_meta)
con_meta <- subset(data, Group == con_code & Condition == cond_meta)
con_out  <- subset(data, Group == con_code & Condition == cond_out)
lib_out  <- subset(data, Group == lib_code & Condition == cond_out)

# Expected credit: the metaperception on the family where each group's
# error was largest
lib_meta$credit <- lib_meta$binding
con_meta$credit <- con_meta$individualizing

# Benchmarks and mismatch scores (for the equivalence check only)
R_lib <- mean(con_out$binding, na.rm = TRUE)
R_con <- mean(lib_out$individualizing, na.rm = TRUE)
lib_meta$mismatch <- R_lib - lib_meta$credit
con_meta$mismatch <- con_meta$credit - R_con
```

2. Descriptives and scale ranges

The observed ranges settle whether the social identity items were on a 5- or 7-point scale (the SI currently says both).

```r
describe <- function(df, label) {
  vars <- c("credit", "identity", "fusion", "ideology")
  data.frame(group = label, variable = vars,
             n = sapply(vars, function(v) sum(!is.na(df[[v]]))),
             M = sapply(vars, function(v) mean(df[[v]], na.rm = TRUE)),
             SD = sapply(vars, function(v) sd(df[[v]], na.rm = TRUE)),
             min = sapply(vars, function(v) min(df[[v]], na.rm = TRUE)),
             max = sapply(vars, function(v) max(df[[v]], na.rm = TRUE)),
             row.names = NULL)
}
kable(rbind(describe(lib_meta, "Liberals"), describe(con_meta, "Conservatives")),
      digits = 2)
```

group	variable	n	M	SD	min	max
Liberals	credit	95	2.00	0.67	1	3.62
Liberals	identity	95	3.83	0.92	1	5.00
Liberals	fusion	95	3.31	0.99	1	5.00
Liberals	ideology	95	2.16	1.35	1	7.00
Conservatives	credit	86	3.06	0.75	2	4.50
Conservatives	identity	86	3.90	0.92	1	5.00
Conservatives	fusion	86	3.50	1.21	1	5.00
Conservatives	ideology	86	5.66	1.60	1	7.00

```r
cat("Identity item range (all participants):",
    range(unlist(data[, identity_items]), na.rm = TRUE), "\n")
```

```
Identity item range (all participants): 1 5
```

```r
cat("Benchmarks: conservatives' actual binding rating of liberals =", round(R_lib, 3),
    "; liberals' actual individualizing rating of conservatives =", round(R_con, 3), "\n")
```

```
Benchmarks: conservatives' actual binding rating of liberals = 2.739 ; liberals' actual individualizing rating
↪  of conservatives = 2.212
```

3. Main models: expected credit ~ fusion + identity + ideology

For liberals, higher expected credit means expecting conservatives to see more binding values in them. For conservatives, it means expecting liberals to see more individualizing values in them.

```r
m_lib <- lm(credit ~ fusion + identity + ideology, data = lib_meta)
m_con <- lm(credit ~ fusion + identity + ideology, data = con_meta)

kable(rbind(fit_summary(m_lib, "Liberals"), fit_summary(m_con, "Conservatives")),
      digits = 3, row.names = FALSE, caption = "Model fit")
```

Table 1: Model fit

model	n	F	df1	df2	p	R2	adj_R2
Liberals	95	13.334	3	91	< .001	0.305	0.282
Conservatives	86	1.885	3	82	.139	0.065	0.030

```r
kable(rbind(coef_table(m_lib, "Liberals"), coef_table(m_con, "Conservatives")),
      digits = 3, caption = "Coefficients (Type III partial eta squared)")
```

Table 2: Coefficients (Type III partial eta squared)

model	term	b	SE	CI_low	CI_high	beta	t	df	p	eta2_p
Liberals	fusion	0.131	0.079	-0.026	0.288	0.193	1.655	91	.101	0.029
Liberals	identity	0.292	0.106	0.081	0.503	0.399	2.755	91	.007	0.077
Liberals	ideology	0.381	0.062	0.258	0.505	0.763	6.143	91	< .001	0.293
Conservatives	fusion	-0.164	0.078	-0.320	-0.009	-0.263	-2.107	82	.038	0.051
Conservatives	identity	0.095	0.117	-0.138	0.329	0.116	0.813	82	.419	0.008
Conservatives	ideology	-0.043	0.062	-0.167	0.081	-0.090	-0.682	82	.497	0.006

Collinearity

```
cat("Fusion-identity correlation, liberals:",
    round(cor(lib_meta$fusion, lib_meta$identity, use = "complete.obs"), 3), "\n")
```

```
Fusion-identity correlation, liberals: 0.661
```

```
cat("Fusion-identity correlation, conservatives:",
    round(cor(con_meta$fusion, con_meta$identity, use = "complete.obs"), 3), "\n\n")
```

```
Fusion-identity correlation, conservatives: 0.51
```

```
kable(data.frame(Liberals = vif(m_lib), Conservatives = vif(m_con)), digits = 2,
      caption = "Variance inflation factors")
```

Table 3: Variance inflation factors

	Liberals	Conservatives
fusion	1.78	1.36
identity	2.75	1.79
ideology	2.02	1.53

4. Check: same model as the mismatch version

Fit and $|t|$ must match. For liberals every slope flips sign; for conservatives nothing changes except the intercept.

```
m_lib_mm <- lm(mismatch ~ fusion + identity + ideology, data = lib_meta)
m_con_mm <- lm(mismatch ~ fusion + identity + ideology, data = con_meta)

eq <- data.frame(
  group = rep(c("Liberals", "Conservatives"), each = 3),
  term  = rep(c("fusion", "identity", "ideology"), 2),
  b_credit   = c(coef(m_lib)[-1], coef(m_con)[-1]),
  b_mismatch = c(coef(m_lib_mm)[-1], coef(m_con_mm)[-1]),
  R2_credit   = rep(c(summary(m_lib)$r.squared, summary(m_con)$r.squared), each = 3),
  R2_mismatch = rep(c(summary(m_lib_mm)$r.squared, summary(m_con_mm)$r.squared), each = 3)
)
kable(eq, digits = 3)
```

group	term	b_credit	b_mismatch	R2_credit	R2_mismatch
Liberals	fusion	0.131	-0.131	0.305	0.305
Liberals	identity	0.292	-0.292	0.305	0.305
Liberals	ideology	0.381	-0.381	0.305	0.305

group	term	b_credit	b_mismatch	R2_credit	R2_mismatch
Conservatives	fusion	-0.164	-0.164	0.065	0.065
Conservatives	identity	0.095	0.095	0.065	0.065
Conservatives	ideology	-0.043	-0.043	0.065	0.065

```
stopifnot(isTRUE(all.equal(eq$R2_credit, eq$R2_mismatch)))
```

5. Single-predictor models

Because fusion and social identity overlap, each is also entered alone (with ideology). A predictor that is significant alone but not in the joint model shares its explanatory variance with the other.

```
single <- rbind(
  coef_table(lm(credit ~ fusion + ideology, data = lib_meta), "Liberals: fusion only"),
  coef_table(lm(credit ~ identity + ideology, data = lib_meta), "Liberals: identity only"),
  coef_table(lm(credit ~ fusion + ideology, data = con_meta), "Conservatives: fusion only"),
  coef_table(lm(credit ~ identity + ideology, data = con_meta), "Conservatives: identity only")
)
kable(single[single$term != "ideology", ], digits = 3, row.names = FALSE)
```

model	term	b	SE	CI_low	CI_high	beta	t	df	p	eta2_p
Liberals: fusion only	fusion	0.244	0.070	0.104	0.383	0.358	3.466	92	< .001	0.116
Liberals: identity only	identity	0.382	0.092	0.200	0.565	0.523	4.171	92	< .001	0.159
Conservatives: fusion only	fusion	-0.139	0.072	-0.282	0.003	-0.223	-1.948	83	.055	0.044
Conservatives: identity only	identity	-0.002	0.110	-0.221	0.217	-0.002	-0.018	83	.986	0.000

6. Comparison with the original specification

The original script: mismatch outcome, proportionality scored as individualizing, Type I η^2. Shown so you can see which reported numbers change and why.

```
old_bind <- c("loyalty_1", "loyalty_2", "authority_1", "authority_2",
              "purity_1", "purity_2")
old_ind  <- c("care_1", "care_2", "equality_1", "equality_2",
              "proportionality_1", "proportionality_2")
lm_old <- lib_meta; cm_old <- con_meta
lm_old$mm <- mean(rowMeans(con_out[, old_bind], na.rm = TRUE)) -
  rowMeans(lm_old[, old_bind], na.rm = TRUE)
cm_old$mm <- rowMeans(cm_old[, old_ind], na.rm = TRUE) -
  mean(rowMeans(lib_out[, old_ind], na.rm = TRUE))
o_lib <- lm(mm ~ fusion + identity + ideology, data = lm_old)
o_con <- lm(mm ~ fusion + identity + ideology, data = cm_old)

kable(rbind(fit_summary(o_lib, "Liberals (original)"),
            fit_summary(o_con, "Conservatives (original)")),
      digits = 3, row.names = FALSE)
```

model	n	F	df1	df2	p	R2	adj_R2
Liberals (original)	95	11.212	3	91	< .001	0.270	0.246
Conservatives (original)	86	1.705	3	82	.172	0.059	0.024

```
kable(rbind(coef_table(o_lib, "Liberals (original)"),
            coef_table(o_con, "Conservatives (original)")),
      digits = 3, caption = "Original specification, with Type III eta2")
```

Table 4: Original specification, with Type III eta2

model	term	b	SE	CI_low	CI_high	beta	t	df	p	eta2_p
Liberals (original)	fusion	-0.095	0.090	-0.273	0.083	-0.126	-1.055	91	.294	0.012
Liberals (original)	identity	-0.304	0.120	-0.542	-0.066	-0.377	-2.538	91	.013	0.066
Liberals (original)	ideology	-0.400	0.070	-0.539	-0.260	-0.725	-5.695	91	< .001	0.263
Conservatives (original)	fusion	-0.109	0.060	-0.228	0.009	-0.229	-1.830	82	.071	0.039
Conservatives (original)	identity	0.052	0.090	-0.126	0.231	0.084	0.585	82	.560	0.004
Conservatives (original)	ideology	-0.039	0.048	-0.133	0.056	-0.107	-0.810	82	.420	0.008

```
kable(rbind(cbind(model = "Liberals (original)",
                  as.data.frame(eta_squared(o_lib, partial = TRUE))[, 1:2]),
            cbind(model = "Conservatives (original)",
                  as.data.frame(eta_squared(o_con, partial = TRUE))[, 1:2])),
      digits = 3, caption = "Original Type I partial eta2 (as reported in the SI)")
```

Table 5: Original Type I partial eta2 (as reported in the SI)

model	Parameter	Eta2_partial
Liberals (original)	fusion	0.001
Liberals (original)	identity	0.012
Liberals (original)	ideology	0.263
Conservatives (original)	fusion	0.051
Conservatives (original)	identity	0.001
Conservatives (original)	ideology	0.008

7. Manuscript-ready numbers

```
write_model <- function(m, label) {
  fs <- fit_summary(m, label)
  ct <- coef_table(m, label)
  cat(sprintf("**%s.** Model: $F(%d, %d) = %.2f$, $p %s$, $R^2 = %s$, adjusted $R^2 = %s$.\n\n",
              label, fs$df1, fs$df2, fs$F,
              ifelse(fs$p == "< .001", "< .001", paste("=", fs$p)),
              sub("^0", "", sprintf("%.2f", fs$R2)),
              sub("^0", "", sprintf("%.3f", fs$adj_R2))))
  for (i in seq_len(nrow(ct))) {
    r <- ct[i, ]
    cat(sprintf("- %s: `$b = %.2f$, SE $= %.2f$, $\\beta = %.2f$, $t(%d) = %.2f$, $p %s$, 95\\%% CI $[%.2f,
     ↪  %.2f]$, $\\eta^2_p = %s$`\n",
                r$term, r$b, r$SE, r$beta, r$df, r$t,
                ifelse(r$p == "< .001", "< .001", paste("=", r$p)),
                r$CI_low, r$CI_high, sub("^0", "", sprintf("%.2f", r$eta2_p))))
  }
  cat("\n")
}
write_model(m_lib, "Liberals")
```

Liberals. Model: $F(3, 91) = 13.33$, $p < .001$, $R^2 = .31$, adjusted $R^2 = .282$.

- fusion: `$b = 0.13$, SE $= 0.08$, $\beta = 0.19$, $t(91) = 1.65$, $p = .101$, 95\% CI $[-0.03, 0.29]$, $\eta^2_p = .03$`
- identity: `$b = 0.29$, SE $= 0.11$, $\beta = 0.40$, $t(91) = 2.75$, $p = .007$, 95\% CI $[0.08, 0.50]$, $\eta^2_p = .08$`
- ideology: `$b = 0.38$, SE $= 0.06$, $\beta = 0.76$, $t(91) = 6.14$, $p < .001$, 95\% CI $[0.26, 0.50]$, $\eta^2_p = .29$`

```
write_model(m_con, "Conservatives")
```

Conservatives. Model: $F(3, 82) = 1.88$, $p = .139$, $R^2 = .06$, adjusted $R^2 = .030$.

- fusion: `$b = -0.16$, SE $= 0.08$, $\beta = -0.26$, $t(82) = -2.11$, $p = .038$, 95\% CI $[-0.32, -0.01]$, $\eta^2_p = .05$`
- identity: `$b = 0.10$, SE $= 0.12$, $\beta = 0.12$, $t(82) = 0.81$, $p = .419$, 95\% CI $[-0.14, 0.33]$, $\eta^2_p = .01$`
- ideology: `$b = -0.04$, SE $= 0.06$, $\beta = -0.09$, $t(82) = -0.68$, $p = .497$, 95\% CI $[-0.17, 0.08]$, $\eta^2_p = .01$`

Session info

```
sessionInfo()
```

```
R version 4.6.0 (2026-04-24)
Platform: aarch64-apple-darwin23
Running under: macOS Tahoe 26.5.2

Matrix products: default
BLAS:   /Library/Frameworks/R.framework/Versions/4.6/Resources/lib/libRblas.0.dylib
LAPACK: /Library/Frameworks/R.framework/Versions/4.6/Resources/lib/libRlapack.dylib;  LAPACK version 3.12.1

locale:
[1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8

time zone: America/Los_Angeles
tzcode source: internal

attached base packages:
[1] stats     graphics  grDevices utils     datasets  methods   base

other attached packages:
[1] knitr_1.51       effectsize_1.0.2 car_3.1-5        carData_3.0-6
[5] broom_1.0.13     readxl_1.5.0.1

loaded via a namespace (and not attached):
 [1] jsonlite_2.0.0     dplyr_1.2.1        compiler_4.6.0     tidyselect_1.2.1
 [5] tidyr_1.3.2        yaml_2.3.12        fastmap_1.2.0      R6_2.6.1
 [9] generics_0.1.4     Formula_1.2-5      backports_1.5.1    tibble_3.3.1
[13] insight_1.5.2      pillar_1.11.1      rlang_1.2.0        xfun_0.59
[17] parameters_0.29.2  datawizard_1.3.1   otel_0.2.0         estimability_2.0.0
[21] cli_3.6.6          magrittr_2.0.5     emmeans_2.0.3      digest_0.6.39
[25] mvtnorm_1.4-1      rstudioapi_0.19.0  lifecycle_1.0.5    vctrs_0.7.3
[29] evaluate_1.0.5     glue_1.8.1         cellranger_1.1.0   abind_1.4-8
[33] bayestestR_0.18.1  rmarkdown_2.31     purrr_1.2.2        tools_4.6.0
[37] pkgconfig_2.0.3    htmltools_0.5.9
```